TIME-OPTIMAL AO-RRT FOR SIMULATED FORMULA E CAR

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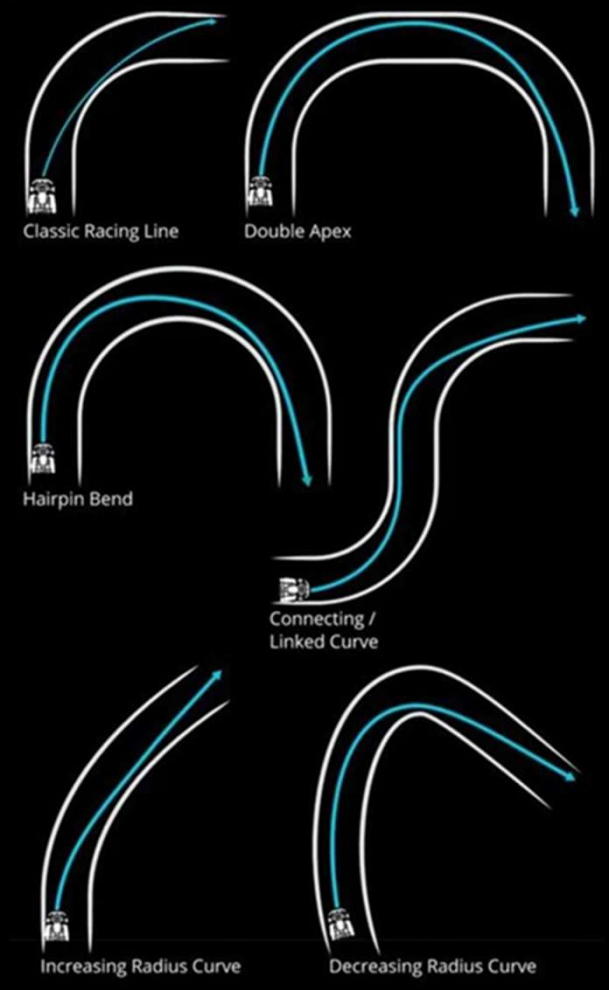
PROJECT OVERVIEW

- Project Goal: **Find optimal race line for known track map**
  - Minimize lap time
  - Respect physics
- Use Sample-Based Planning
  - Large, continuous state-space
  - Complex vehicle dynamics
- Use Indianapolis Motor Speedway as testbed
  - Generalize to other tracks



AO-RRT REVIEW

1) Sample

$x_{rand}, c_{rand}, t_{rand}, u_{rand}$

2) Find Nearest

x_{rand}, c_{rand}

3) Propagate

$x_{near}, u_{rand}, t_{rand}$

4) Trajectory Cost

$c_{new} = cost(y_{near}) + cost(\pi_{new})$

5) Grow Tree

If feasible

6) Update Min

If $x_{new} \in X_g$ and $c_{new} < c_{max}$

WHY WE CHOSE AO-RRT

- Sampling – no discretization of large map
- Handles kinodynamic constraints
- Asymptotically optimal
- Allows custom cost function

VEHICLE MODEL

Friction Circle:

- $m_{car}\sqrt{a_t^2 + a_l^2} \leq \mu_{tires}\, f_n(m, v)$
- Normal force includes downforce (function of velocity)
  - Drag and downforce taken from CFD Analysis [1]
- Track assumed perfectly flat
- $\mu = 1.5$ – Consistent with performance road tire (not slick)

Bicycle Motion Model:

- State
  - (x, y) – position of rear tire
  - θ – orientation about z-axis
  - Added:
    - $S(x, y)$ – track centerline progress
    - v – forward velocity *(added for dynamics)*
- Controls/Inputs
  - a – Throttle/Brake
  - ψ – Steering Angle

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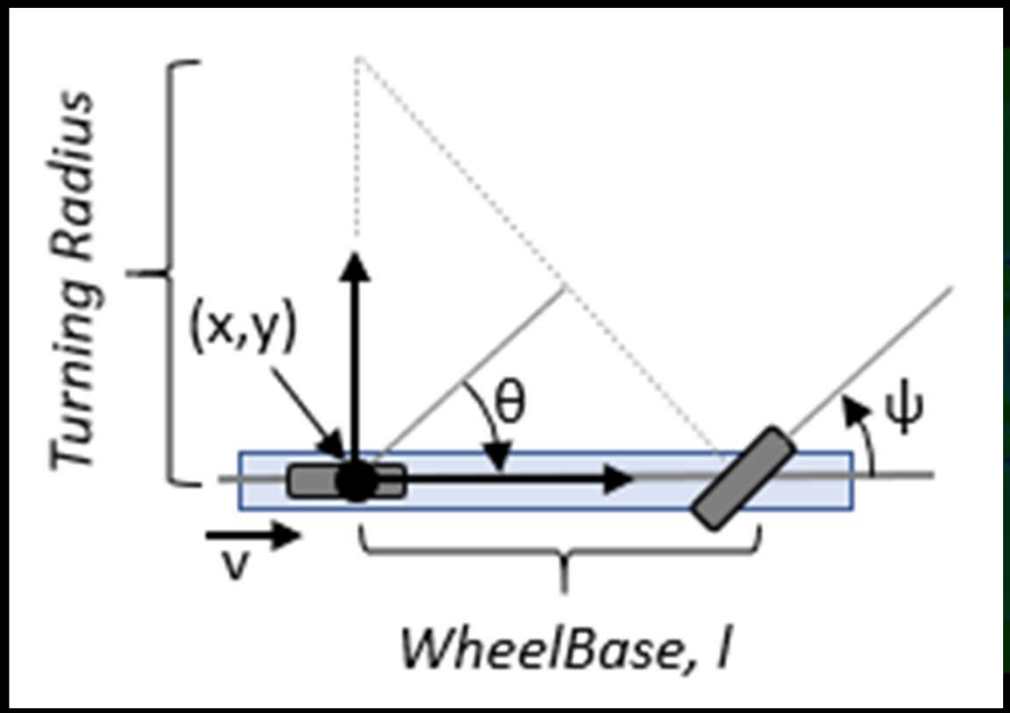

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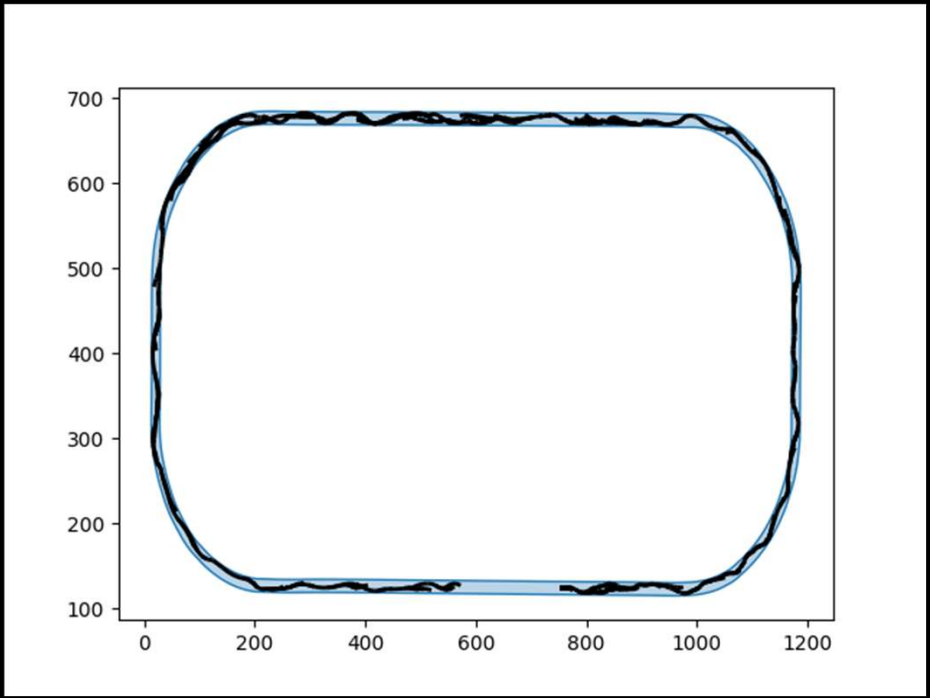
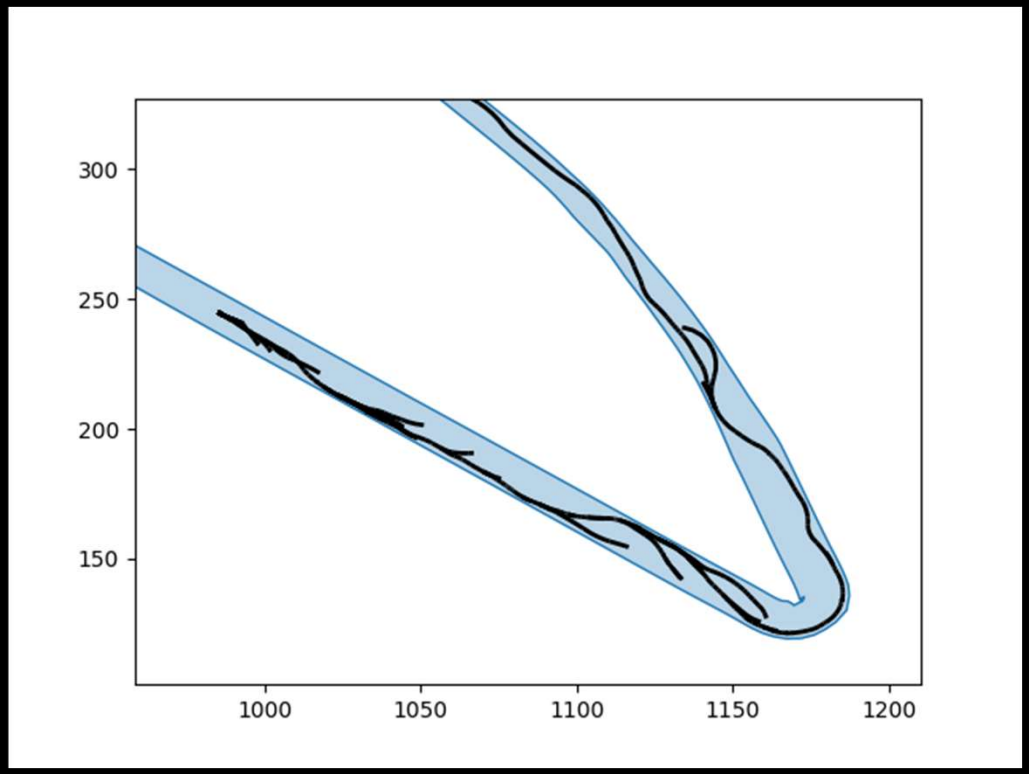
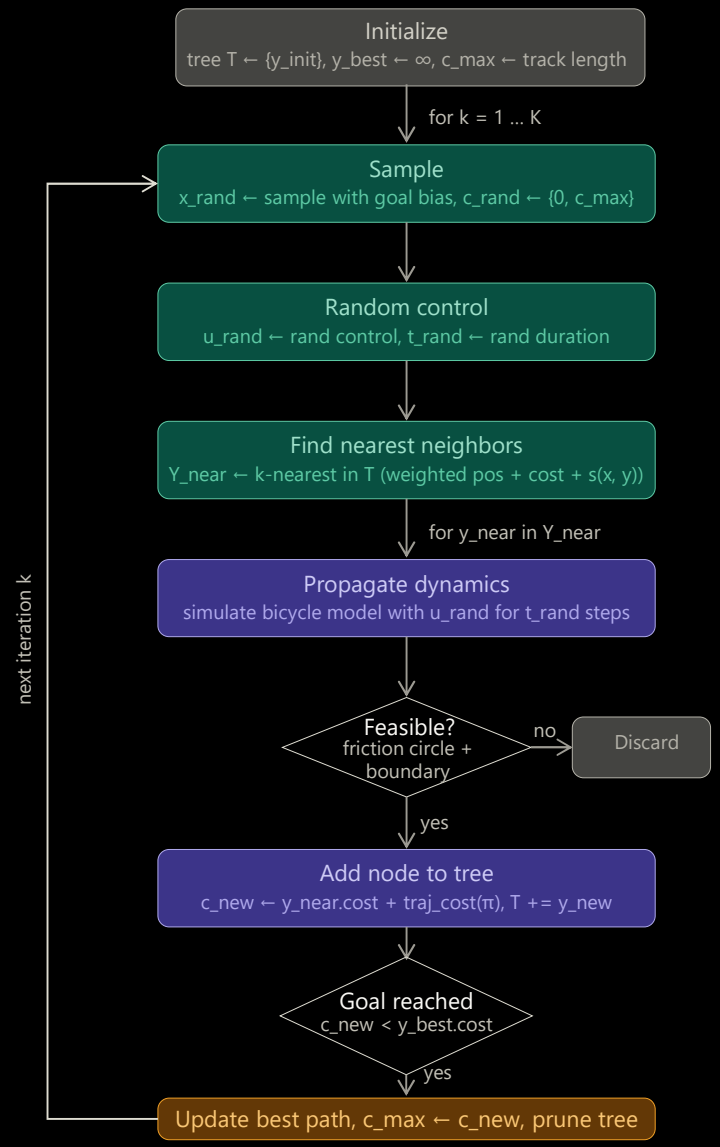
AO-RRT MODIFICATIONS

- Gaussian Control Sampling
  - Usually you're driving (mostly) straight.
  - Usually you want to be on the gas OR on the brake
  - Still probabilistically complete
- Track Centerline Progress Parameterization $S(x, y)$
  - Goal samples attach to root node
  - Cost Function: $\mathrm{T} = \int_0^L \frac{1}{v(s)} ds$
- Collision Checking Includes Tire Slip
- K-Nearest Neighbor Expansion
  - Helps get out of terminal states



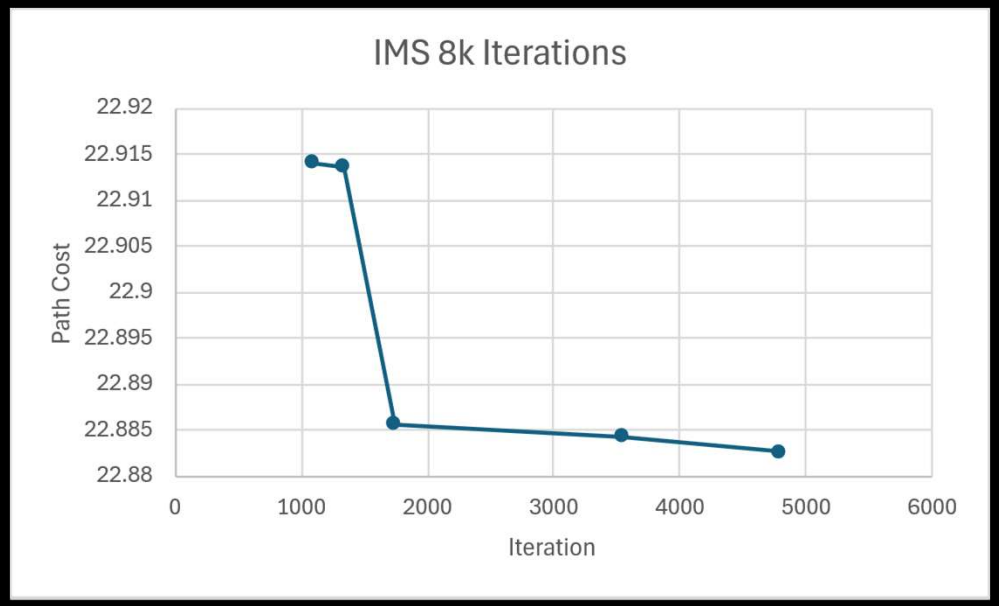
AO-RRT PSEUDOCODE





RESULTS – ASYMPTOTIC OPTIMALITY

- Segments close to the finish are optimized early on
- Early mistakes take extremely long to improve
- This run took just over 3 hours
  - No significant improvement for 40k iterations
- As $t \to \infty$, we approach the optimal path.
  - It might have to be really close to $t = \infty$ ...



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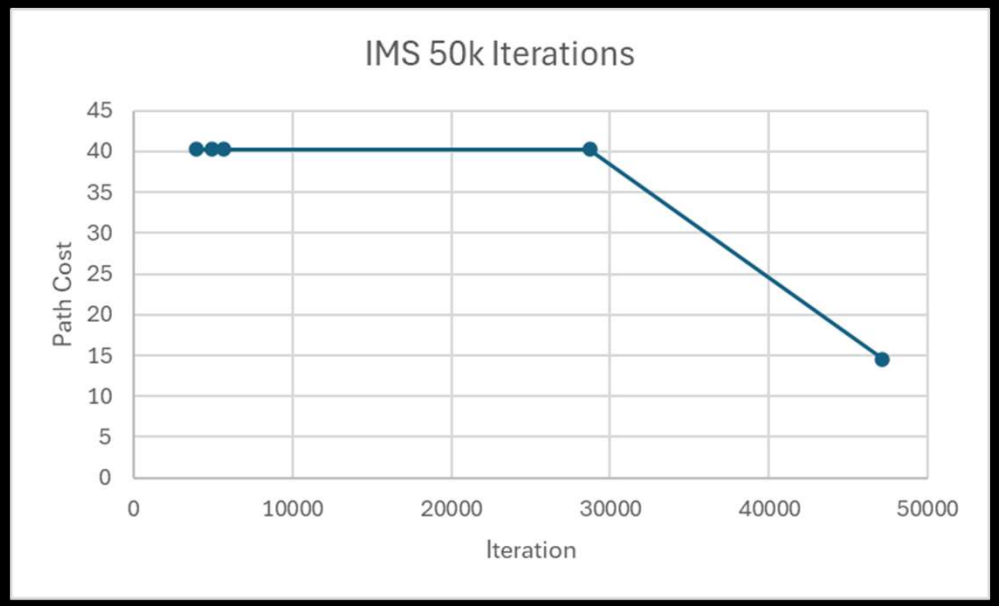

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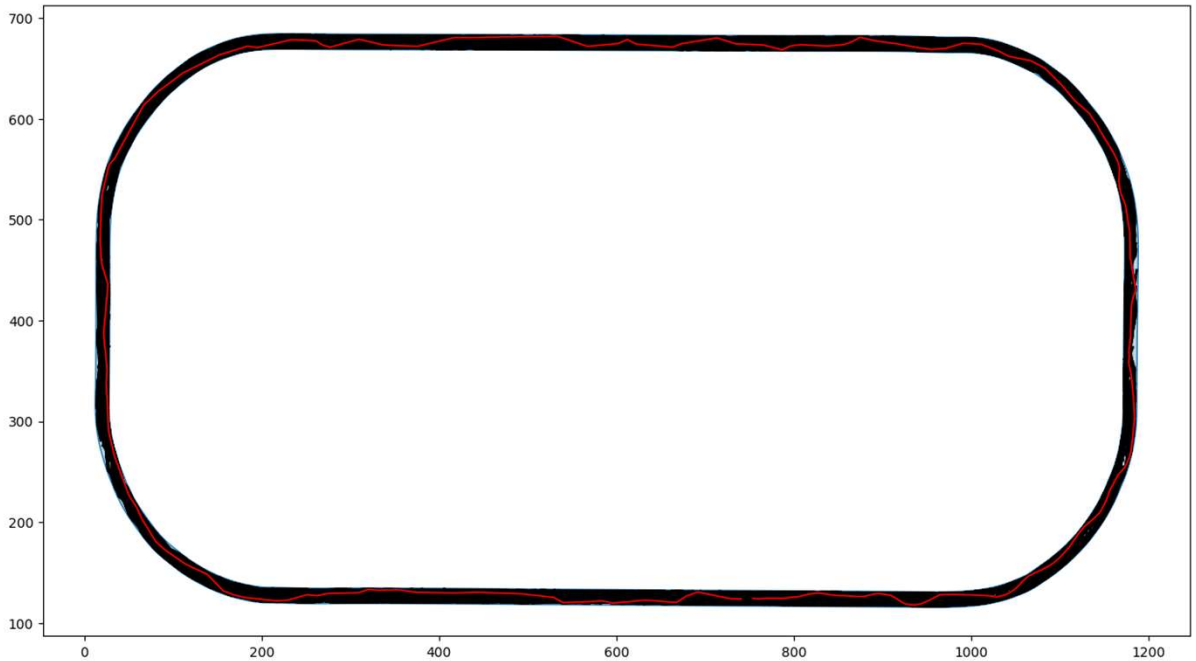
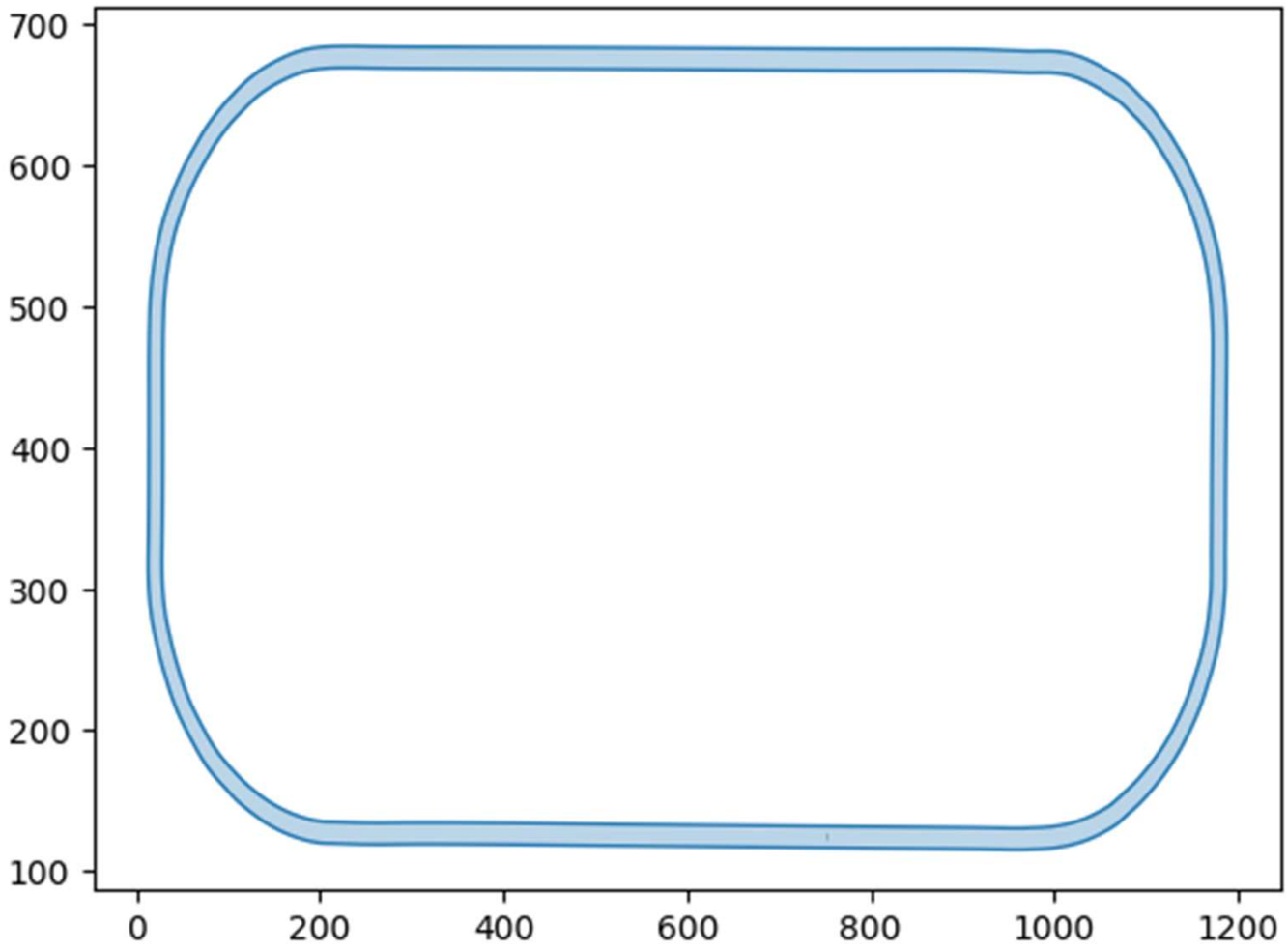


RESULTS: GENERATED PATHS



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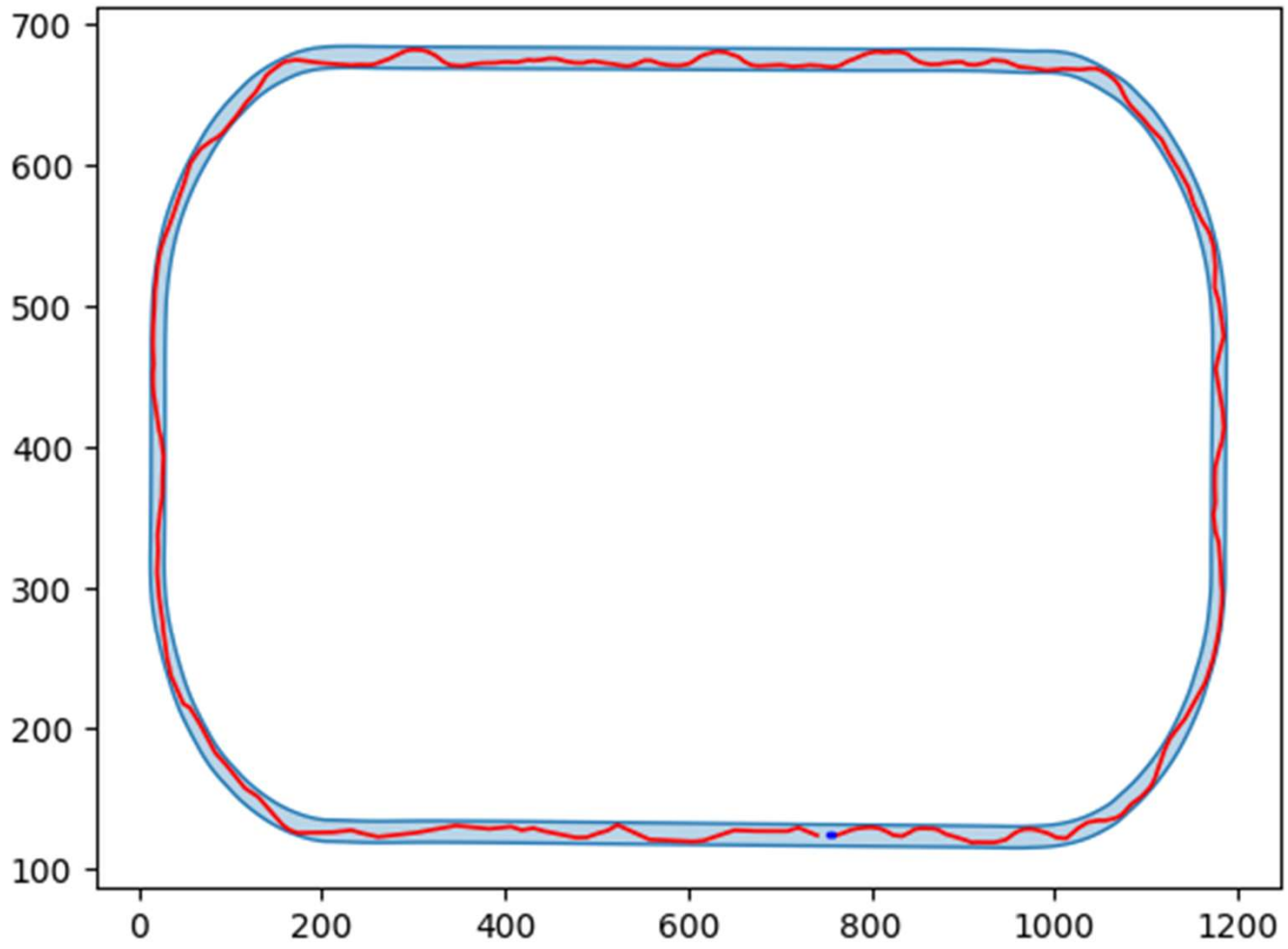
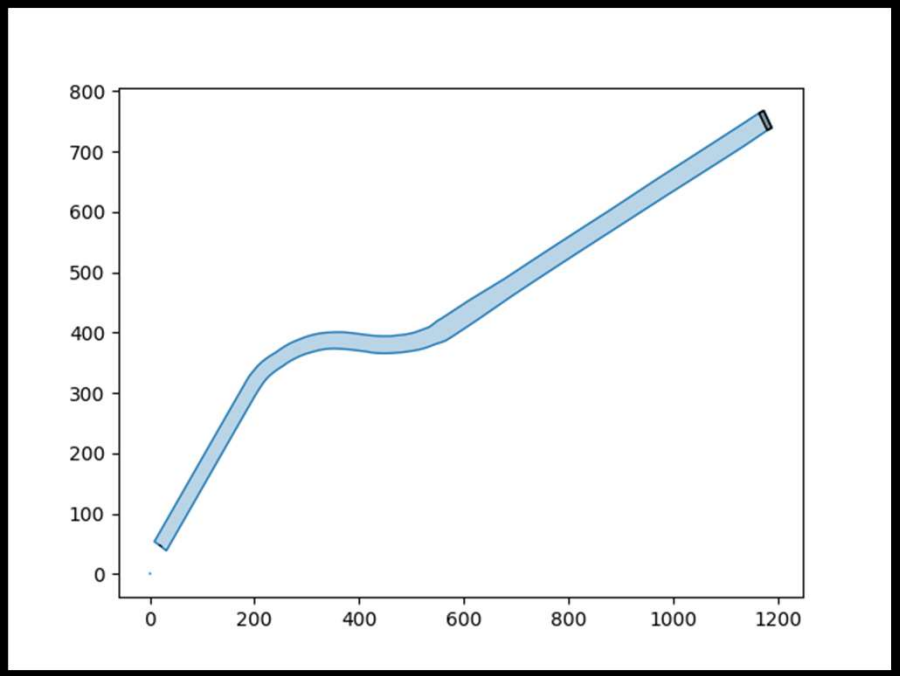
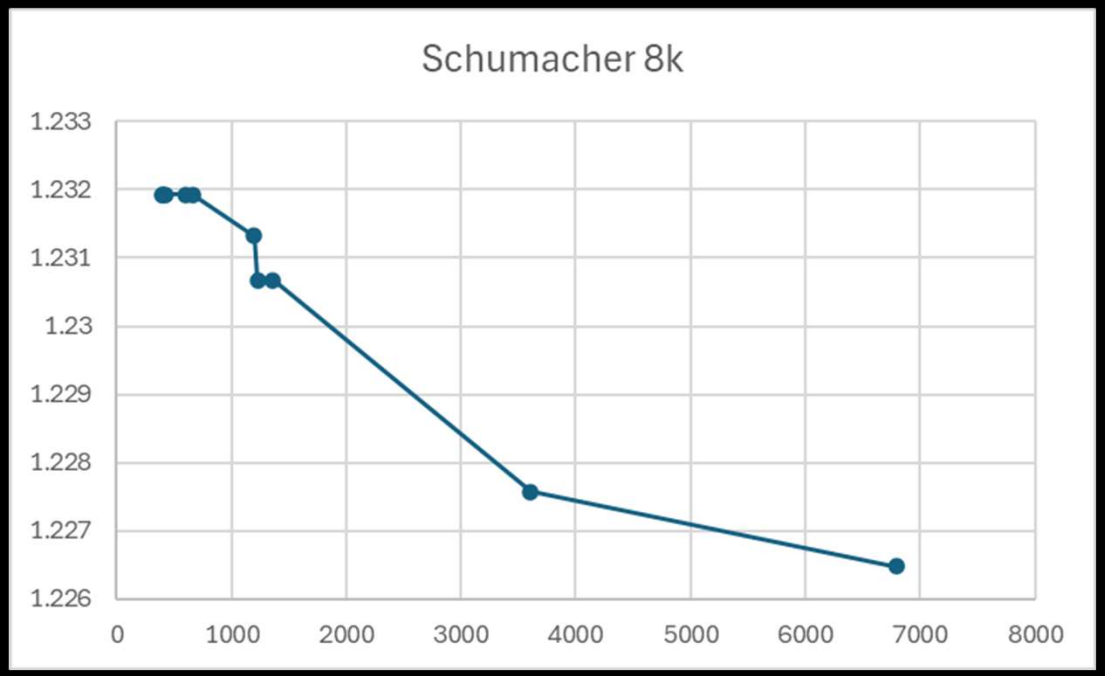


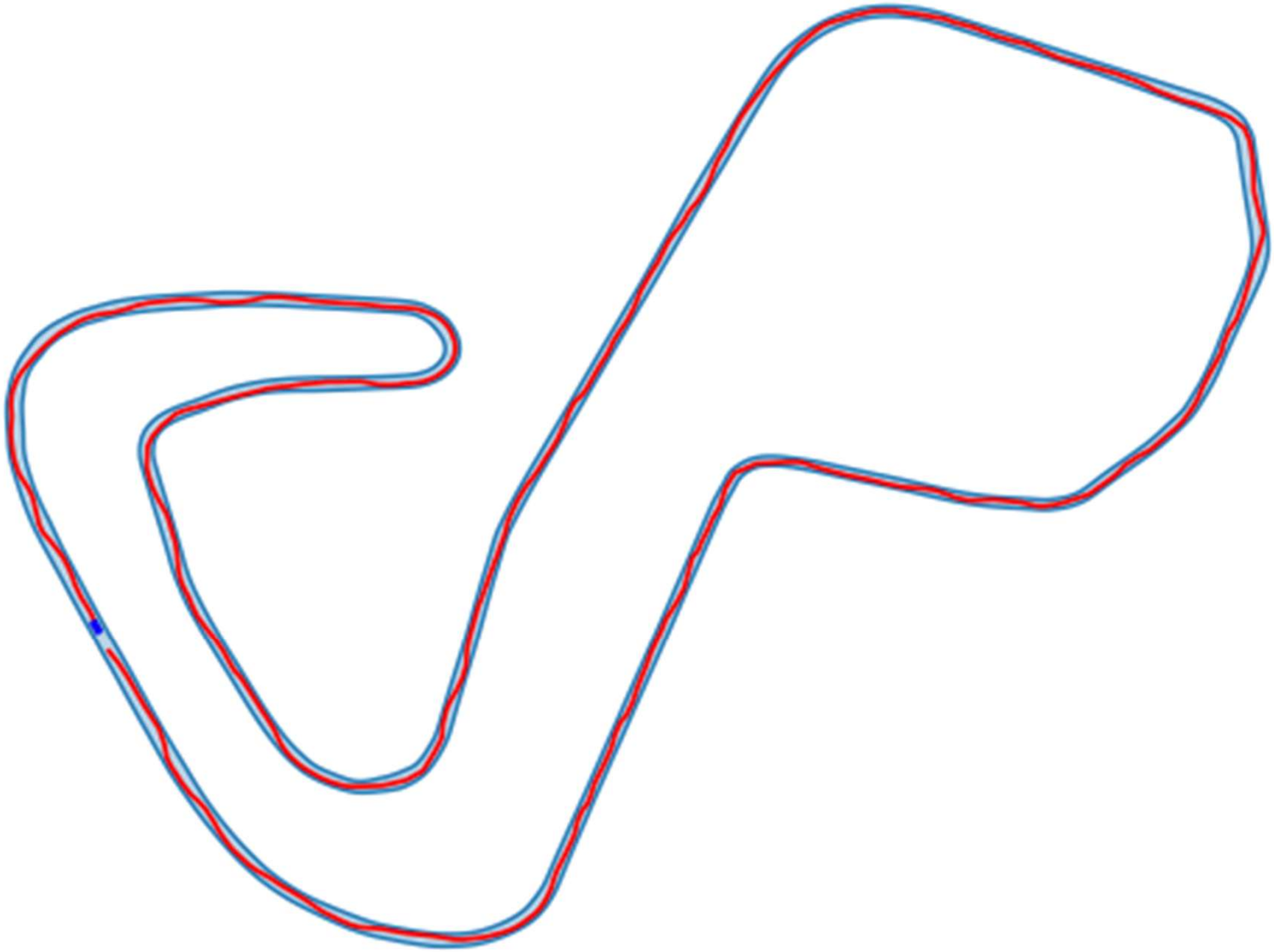
WHAT ABOUT FOR A SHORT TRACK SEGMENT





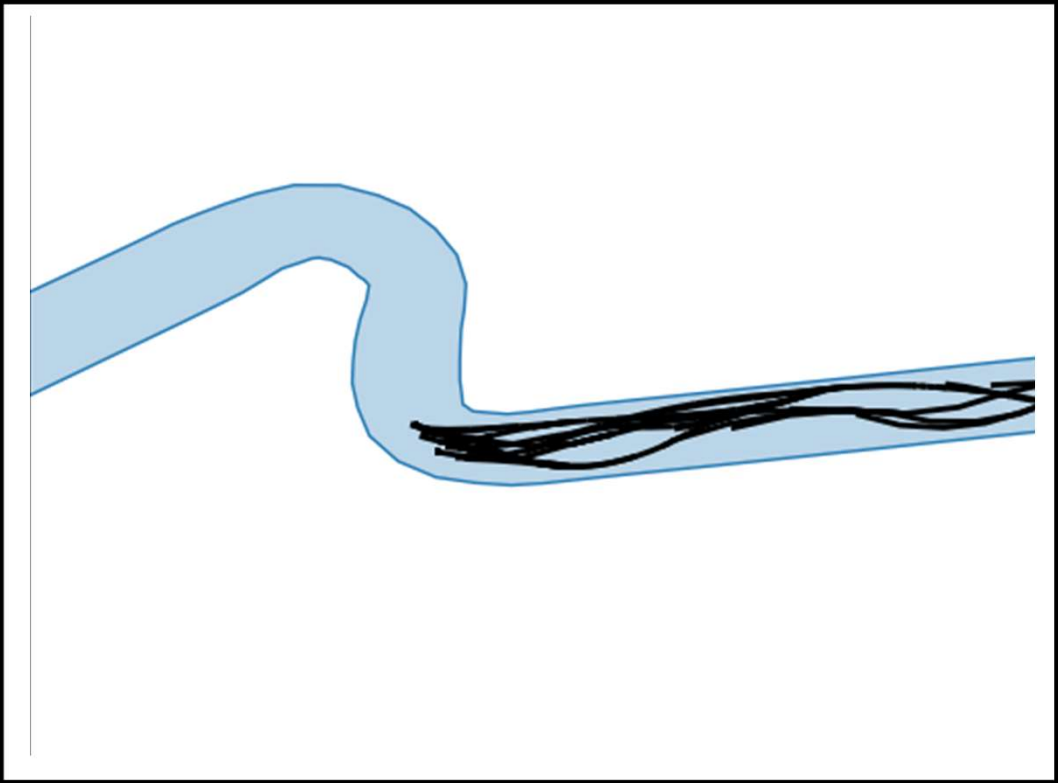
OTHER TRACKS

BRANDS HATCH



PROJECT LIMITATIONS

- AO-RRT is limited for large tracks
- Simple physics model
  - Lap times not comparable to real life
- Very short rollouts
- Instantaneous change in steering angle ψ
- Track Elevation/Banking neglected



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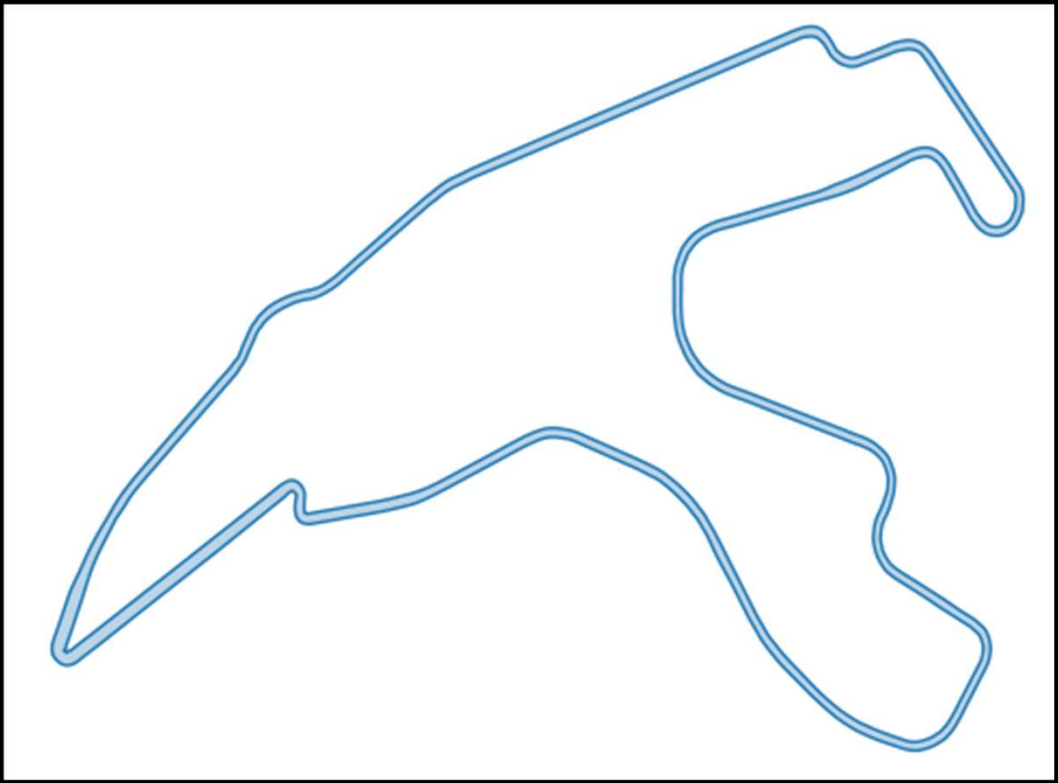
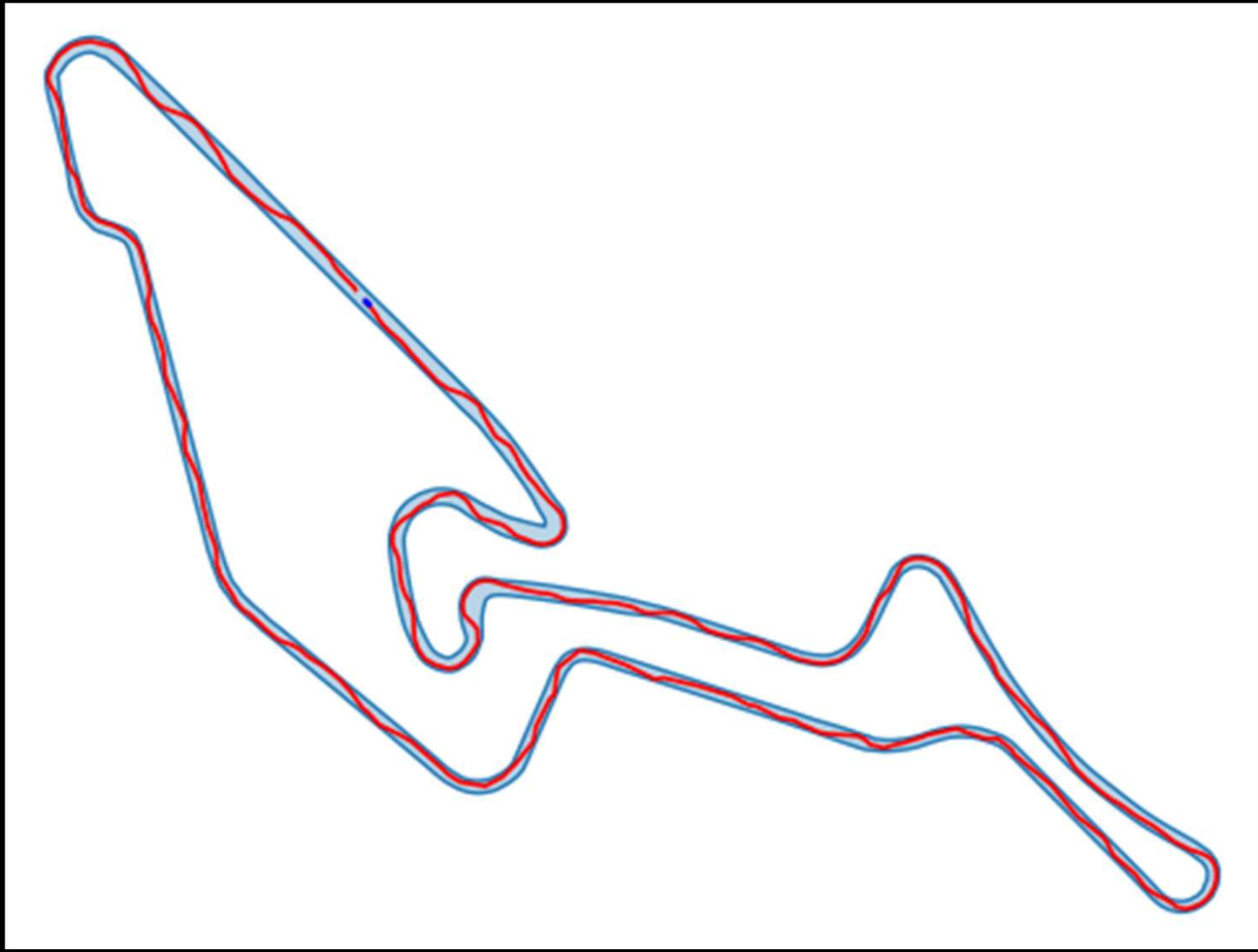
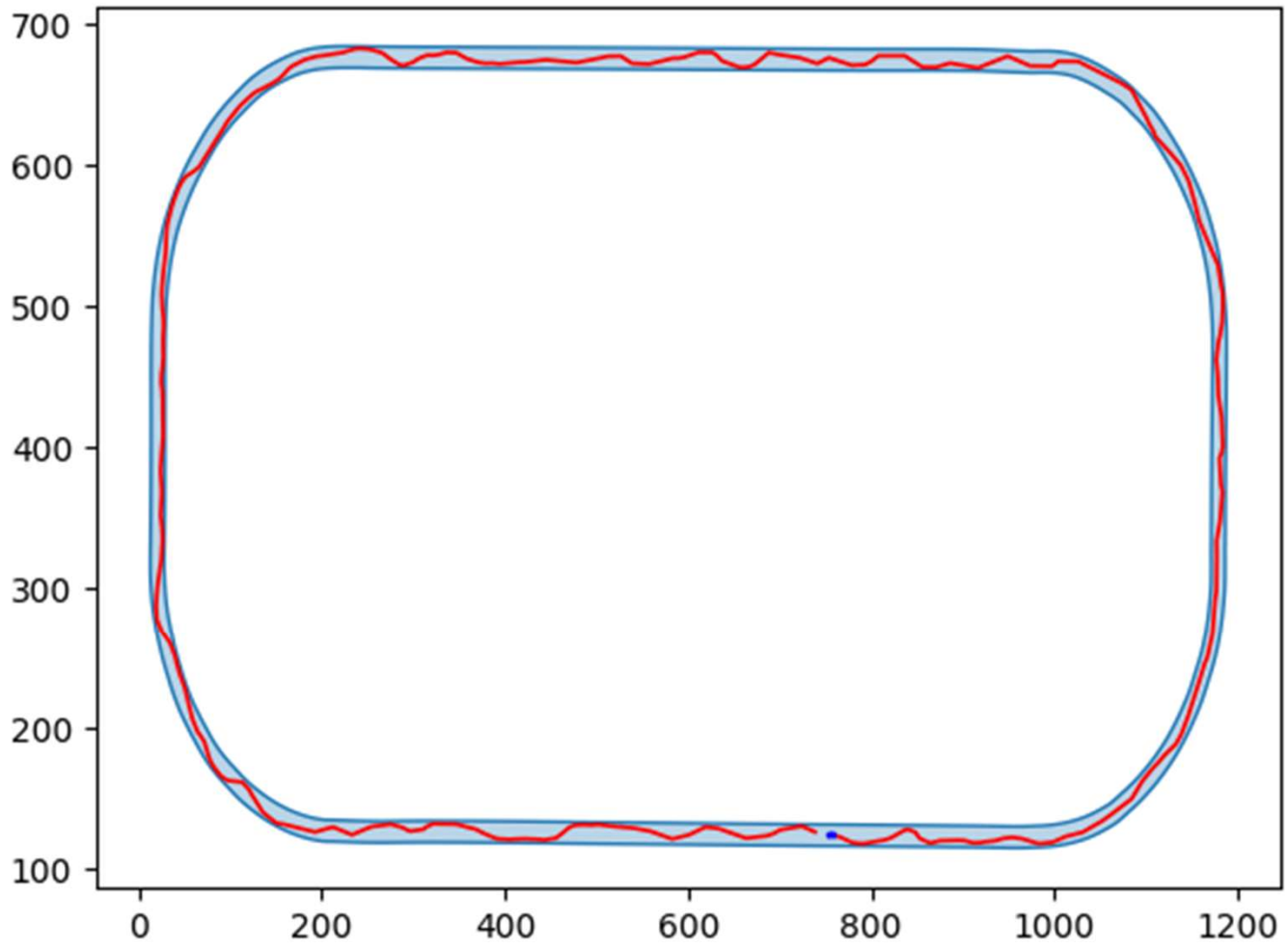


FUTURE WORK

- Optimizing data structure (KD tree or similar)
- Further Parameter tuning
- Heuristic pruning
- Use a different algorithm (MPPI, perhaps)
- Inject further racing knowledge
  - Cost function
  - Guided Sampling
- Plan for track segments (i.e. a single corner or series of corners)

QUESTIONS?



[1] AirShaper Analysis by Wouter Remmerie:
https://www.linkedin.com/pulse/aero-shake-down-formula-e-versus-one-wouter-remmerie

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